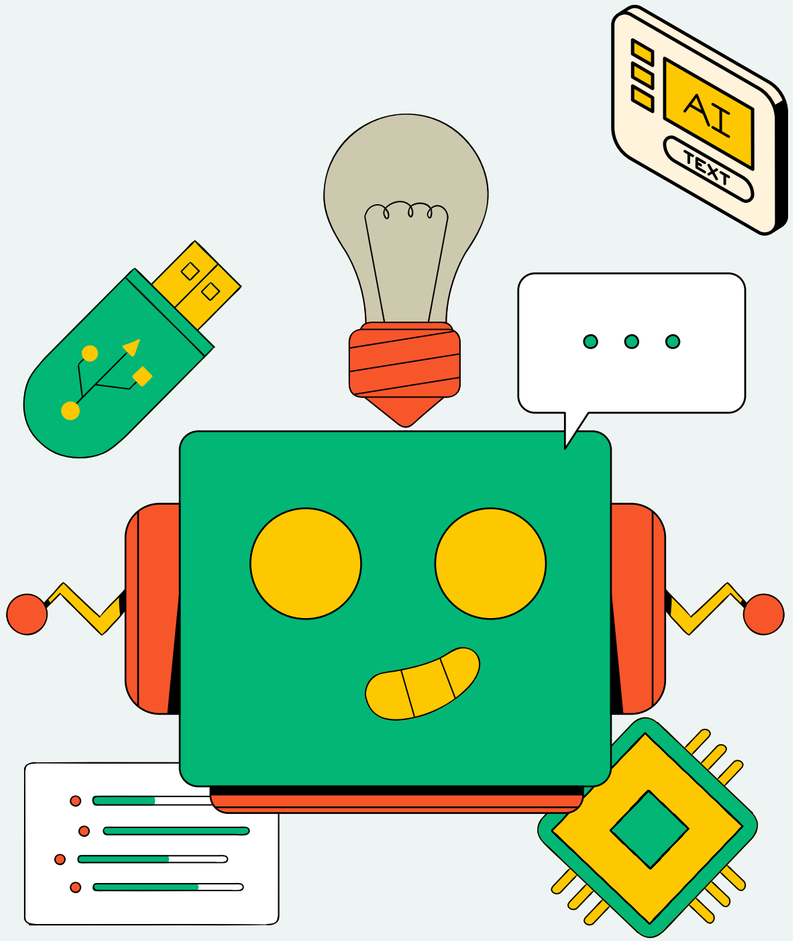
THYNK UNLIMITED

WE LEARN FOR THE FUTURE

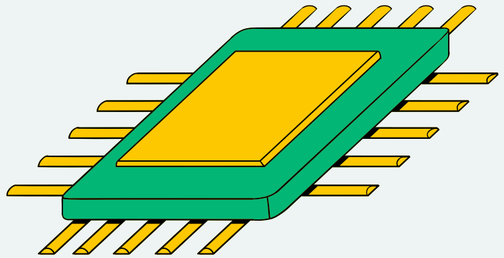


STARCAST AI

MOVIE BOX OFFICE REVENUE PREDICTOR

PRESENTED BY:

TEAM ALPHA





PRESENTATION OUTLINE

- Project Overview
- Key Concepts
- Key Features
- System Architecture
- The Box Office Dashboard
- Behind the Scenes
- Supporting Cast & Integrations
- The Good & The Gaps
- Conclusion

PROJECT OVERVIEW

The Movie Box Office Revenue Predictor is a next-generation full-stack application designed to estimate a film's domestic box office performance using advanced machine learning.



This system empowers analysts, film students, producers, researchers, and creators to explore how movie attributes influence commercial outcomes — before the movie is even released.

KEY CONCEPTS

Data-Driven Decision Support

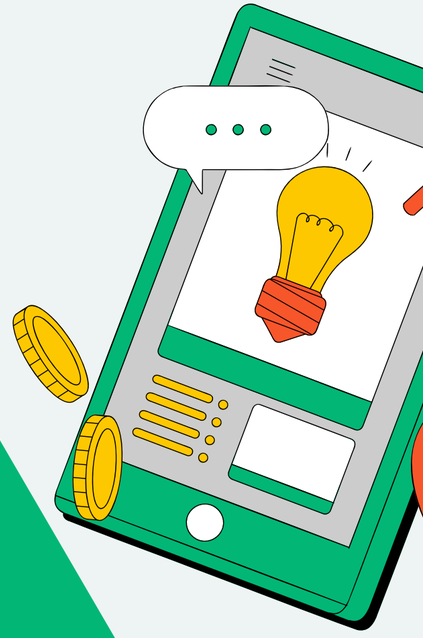
Machine Learning Regression

ROI & Financial Metrics

Real-Time Revenue Prediction

Input Attribute Analysis

Model Interpretability





KEY FEATURES

- Real-time box office revenue prediction using a trained XGBoost model.
- Interactive charts displaying revenue ranges, ROI, and confidence intervals.
- Fast, responsive React + Vite frontend with modern UI styling.
- High-performance FastAPI backend for low-latency ML inference.
- End-to-end type-safe architecture powered by TypeScript and Pydantic.



SYSTEM ARCHITECTURE



01

THE BOX OFFICE DASHBOARD

- **React** – Component-driven, dynamic UI
- **TypeScript** – End-to-end type safety
- **Vite** – Ultra-fast dev server + build tool
- **Tailwind CSS** – Modern utility-first styling
- **Dark Mode Layout** – Modern, cinematic feel
- **Responsive UI** – Works across all screens

02

BEHIND THE SCENES

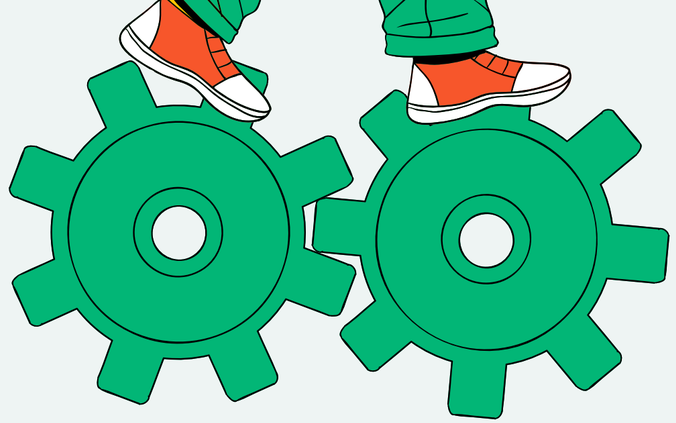
- **FastAPI –** High-performance REST API
- **Python ML Ecosystem**
- **XGBoost –** Primary prediction model
- **Scikit-learn –** Feature processing & utilities
- **Pandas / NumPy –** Data transformations
- **Joblib –** Model loading & serialization

03

SUPPORTING CAST & INTEGRATIONS

- **Axios →** API calls
- **Model Loader →** Load XGBoost via Joblib
- **FastAPI Endpoints →** /predict, /options, /model/info
- **Vite HMR →** Hot UI reload
- **TypeScript Types →** Shared definition for clean interfacing
- **CORS + REST APIs →** Cross-origin communication

THE BOX OFFICE DASHBOARD



REACT

- Component-driven architecture
- Enables dynamic UI rendering
- Excellent performance with concurrent features

TYPESCRIPT

- Strict type-safety for predictable behavior
- Reduces bugs across API interactions
- Improves scalability and team collaboration

VITE

- Lightning-fast development server
- Instant HMR (Hot Module Replacement)
- Highly optimized production bundling

TAILWIND CSS

- Utility-first styling for rapid UI creation
- Consistent, responsive, modern layouts
- Perfect for dark theme + dashboard interfaces

BEHIND THE SCENES

FastAPI

- High-performance async framework
- Auto-generates Swagger & Redoc docs
- Ideal for ML prediction APIs

Scikit-learn

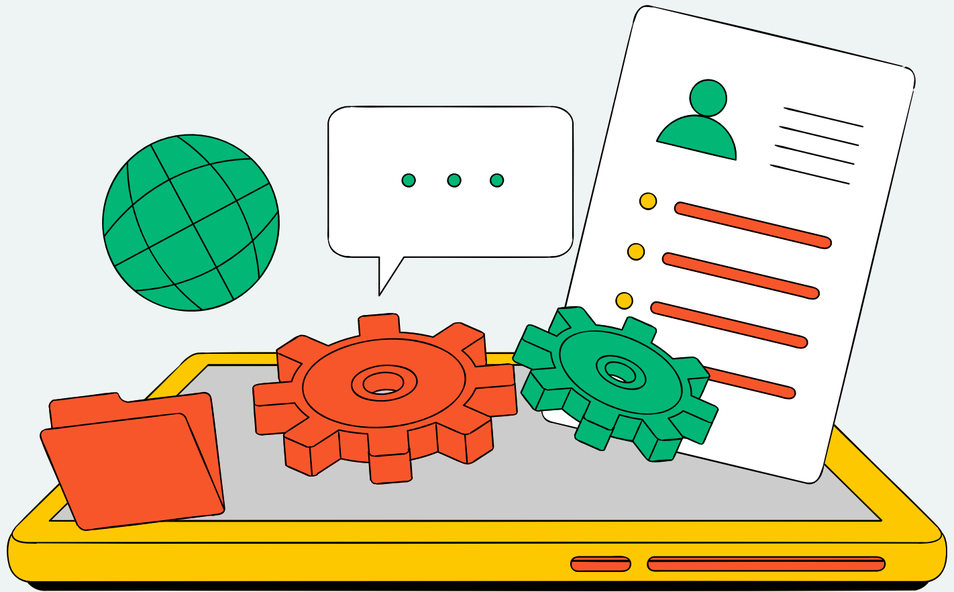
- Feature engineering utilities
- Scaling, preprocessing, transformation helpers

XGBoost

- Industry-leading regression algorithm
- Handles non-linear relationships effectively
- High accuracy for box office prediction tasks

Pandas & NumPy

- Data manipulation & numeric processing
- Essential for handling complex movie datasets



SUPPORTING CAST & INTEGRATIONS

AXIOS

- Manages all frontend-to-backend communication
- Handles response mapping & error states

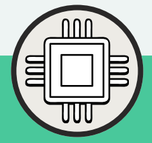
MODEL LOADER

- Loads XGBoost model into memory
- Ensures consistent predictions each request

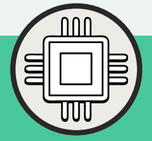
CORS + REST APIS

- Seamless cross-origin communication
- Allows local React frontend to connect cleanly

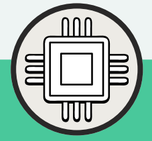
THE GOODS & THE GAPS



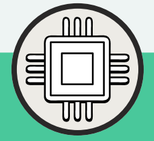
Fast predictions with optimized FastAPI



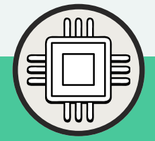
Accurate model with XGBoost



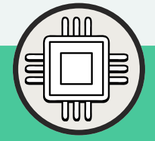
Strong data validation



Model performance tied to real market volatility

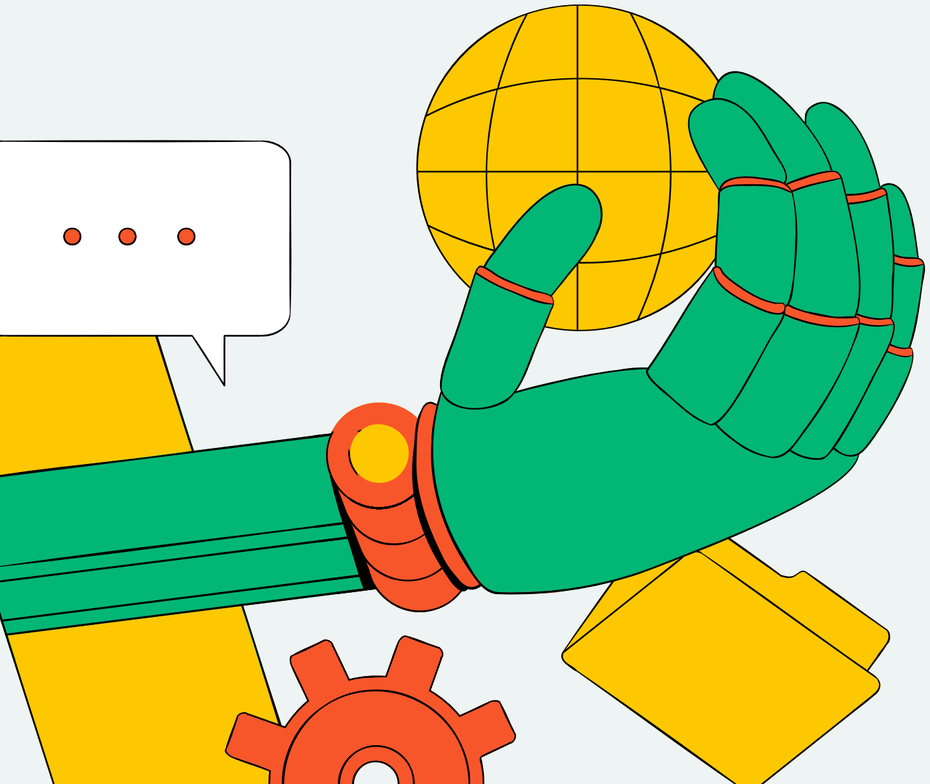


Requires continuous data refresh



Infrastructure costs rise with high traffic

CONCLUSION



StarCast AI successfully brings data-driven intelligence to movie revenue forecasting. The system delivers clear insights such as revenue estimates, ROI, and confidence intervals.
Although real-world challenges like shifting audience trends and limited training data can affect accuracy, the platform sets a strong foundation for scalable, future-ready box office analytics.

THANK YOU

ADITYA RAIZADA

SEJAL BHARDWAJ

